MGU UGP(HONOURS) REGULAR EXAM – OCTOBER 2025

THIRD SEMESTER

MG3DSCPHL200- SYMBOLIC LOGIC

ANSWER KEY

PART A

1. When the conclusion logically follows from the premises
2. a. Universality
   b. precision
   c. clarity
   d. reduces ambiguity

3. A statement that is either true or false

4. p.q is false.

5. Ravi is a boy and Gita is a girl (Any sentence with a conjunction 'and')

6. 1) A.B

   2) (AvC)→ D/ ∴ A.D

   3) A (1, simplification)

   4)A V C (3, addition)

   5) D (2, 4, Modus ponens)

   6. A. D(3, 5, conjunction)

7. p v q or p v (q.r)

8. Any statement that can be either true or false

   Eg. It is hot in April

   Dust causes allergy

9.A rule of inference where conclusion is drawn from an implication and the negation of its consequent

p→q

~ q

∴~p

10. (x)(Cx→~Vx), where Cx,x is child and Vx,x is voter

11. (∃x) (Fx.Rx) where Fx , x is flower and Rx, x is red

12. ∃ (there exists)

Part B

13. Reasoning made simple

    Detects errors easily

    Allows precision and clarity

    Focus on form over content

14.Tradiitonal logic: Based on Aristotelean logic

    Uses ordinary language

    Unit of analysis is terms

    Limited to simple forms of reasoning

    Symbolic logic: Based on modern mathematical logic

    Uses symbols

    Unit of analysis is proposition

    Involves simple and complex reasoning

15.

p	q	p→q
T	T	T
T	F	F
F	T	T
F	F	T

16.a. If p is true,

P. ~P is T. F ie, F

If p is false

Then it is F

b. If p is true,

p V ~y is T v T ie, T

if p is false

F v T ie T

17. Let Alice is president be p

Betty is elected Vice president be q

Carol is treasurer be r

p→(qv r)

q

∴ p→~r

P	q	r	qvr	p→(qv r)	~r	p→~r
T	T	T	T	T	F	F
T	T	F	T	T	T	T
T	F	T	T	T	F	F
T	F	F	F	F	T	T
F	T	T	T	T	F	T
F	T	F	T	T	T	T
F	F	T	T	T	F	T
F	F	F	F	T	T	T

In 1st row, premises are true and conclusion false, therefore argument is invalid

18. a. p→(q.r)

p v(q.r)

∴ q.r

b. pv(q. ~p)

p

∴ ~(q.~p)

19.

p	q	r	~r	q→~r	p→(q→~r)
T	T	T	F	F	F
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T

F	T	T	F	F	T
F	T	F	T	T	T
F	F	T	F	T	T
F	F	F	T	T	T

Since conclusion is not necessarily true , it is not a tautology

b. p	q	r	pvq	~(pvq)	~r	~(pvq) v ~r
T	T	T	T	F	F	F
T	T	F	T	F	T	T
T	F	T	T	F	F	F
T	F	F	T	F	T	T
F	T	T	T	F	F	F
F	T	F	T	F	T	T
F	F	T	F	T	F	T
F	F	F	F	T	T	T

Since conclusion is not necessarily true , it is not a tautology

20. Simple proposition: Expresses a single idea

Contains one variable

Eg.p , q

Complex proposition: Combines 2 or more simple propositions

Contains more than one variable which are connected with \ logical connectives

Eg. p.q, pv r etc

21. a. Ir (I is intelligent, r is radha)

b. Cn (C is capital of india and n is new delhi)

c. Er(E is engineer and r is ravi)

d. Ig. Hg (I is intelligent, H is hardworking and g is Govind)

e. Tc(T is travellor and C is Columbus)

Part C

23 . Propositional logic:

- Treats whole proposition as single unit.
- Lacks Internal analysis.
- Statements involving generality cannot be expressed

Predicate logic

- Internal analysis is possible
- Uses quantifiers to express generality
- Relationship between terms are expressed

24. In propositional logic:

Constants represent whole propositions

Denoted by capital letters

Eg P Q

Variables represent unspecified propositions

It is rarely used

In predicate logic

Constants represent specific individuals or objects

Eg, R, S

Variables represent arbitrary objects

Eg. x, y

25. a.

p	q	r	~p~q	pvq	~p→r	~q.r	(~p→r). (~q.r)	(pvq)→[(~p→r). (~q.r))
T	T	T	F	F	T	T	F	F	F
T	T	F	F	F	T	T	F	F	F

T F T F T T T T T T

T F F F T T T F F F

F T T T F T T F F F

F T F T F T F F F F

F F T T T F T T T T

F F F T T F F F F T

Since conclusion is not necessarily false, it is not a contradiction

b.

p	q	r	p→q	q→r	(p→q). (q→r)	~p ~pvr	[(p→q). (q→r)]→(~pvr)
T	T	T	T	T	T	F T	T
T	T	F	T	F	F	F F	T
T	F	T	F	T	F	F T	T
T	F	F	F	T	F	F F	T
F	T	T	T	T	T	T T	T
F	T	F	T	F	F	T T	T
F	F	T	T	T	T	T T	T
F	F	F	T	T	T	T T	T

Since conclusion is not necessarily false, it is not a contradiction

26. Four types of categorical propositions

A- Universal affirmative represented as (x)(Sx→Px)

E- Universal negative - (x)(Sx→~Px)

I.Particular affirmative- (∃x) (Sx. Px)

O- Particular negative- (∃x)(Sx. ~Px)